

Prayas JEE 2025

Maths

DPP: 2

Sequence and Series

- Q1** If the sum of n terms of an AP is $2n^2 + 5n$, then the n^{th} term will be
 (A) $4n + 3$ (B) $4n + 5$
 (C) $4n + 6$ (D) $4n + 7$
- Q2** If $\frac{3+5+7+\dots+\text{to } n \text{ terms}}{5+8+11+\dots+\text{to } 10 \text{ terms}} = 7$, then the value of n is
 (A) 35 (B) 36
 (C) 37 (D) 40
- Q3** In an A.P., the p^{th} term is $\frac{1}{q}$ and the q^{th} term is $\frac{1}{p}$. Find the $(pq)^{\text{th}}$ term of the A.P.
- Q4** If 100 times the 100^{th} term of an AP with non-zero common difference equals to 50 times its 50^{th} term, then the 150^{th} term of an AP is
 (A) -150
 (B) 150 times its 50^{th} term
 (C) 150
 (D) Zero
- Q5** If the first, second and last terms of an arithmetic series are a, b and c respectively, then the number of terms is
 (A) $\frac{b+c-2a}{b-a}$
 (B) $\frac{b+c+2a}{b-a}$
 (C) $\frac{b+c-2a}{b+a}$
 (D) $\frac{b+c+2a}{b+a}$
- Q6** Let S_n denotes the sum to n terms of an A.P. whose first term is a . If the common difference d is given by $d = S_n - kS_{n-1} + S_{n-2}$, then k is equal to
 (A) 1 (B) 2 (C) 3 (D) 4
- Q7** If a, b, c are in AP, then the value of $(a + 2b - c)(2b + c - a)(a + 2b + c)$ is
 (A) $4abc$
 (B) $8abc$
 (C) $3abc$
 (D) $16abc$
- Q8** If m^{th} terms of the series $63 + 65 + 67 + 69 + \dots$ and $3 + 10 + 17 + 24 + \dots$ is equal, then m is equal to
 (A) 11 (B) 12
 (C) 13 (D) 15
- Q9** If n is even, then the sum of n terms of the series $1 - 2 + 3 - 4 + 5 - 6 + \dots$, will be
 (A) $-\frac{n}{2}$
 (B) $\frac{n-1}{2}$
 (C) $\frac{n+1}{2}$
 (D) $\frac{2n+1}{2}$
- Q10** The p^{th} term of the series $(3 - \frac{1}{n}) + (3 - \frac{2}{n}) + (3 - \frac{3}{n}) + \dots$ will be
 (A) $3 + \frac{p}{n}$
 (B) $3 - \frac{p}{n}$
 (C) $3 + \frac{n}{p}$
 (D) $3 - \frac{n}{p}$
- Q11** If the sum of the 10 terms of an AP is 4 times to the sum of its 5 terms, then the ratio of first term and common difference is
 (A) 1 : 2 (B) 2 : 1
 (C) 2 : 3 (D) 3 : 2



- Q12** If $\frac{a^{n+1}+b^{n+1}}{a^n+b^n}$ is the AM of a and b , then n is equal to
- (A) 1 (B) -1
(C) 0 (D) None of these

- Q13** If $a_1, a_2, a_3, \dots, a_n$ are in AP, where $a_i > 0$ for all i , then the value of

$$\frac{1}{\sqrt{a_1}+\sqrt{a_2}} + \frac{1}{\sqrt{a_2}+\sqrt{a_3}} + \dots + \frac{1}{\sqrt{a_{n-1}}+\sqrt{a_n}}$$

- (A) $\frac{n-1}{\sqrt{a_1}+\sqrt{a_n}}$
(B) $\frac{n+1}{\sqrt{a_1}+\sqrt{a_n}}$
(C) $\frac{n-1}{\sqrt{a_1}-\sqrt{a_n}}$
(D) $\frac{n+1}{\sqrt{a_1}-\sqrt{a_n}}$



Answer Key

Q1 (A)

Q2 (A)

Q3 1

Q4 (D)

Q5 (A)

Q6 (B)

Q7 (D)

Q8 (C)

Q9 (A)

Q10 (B)

Q11 (A)

Q12 (C)

Q13 (A)

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